BTC Carry Fund: Risk Architecture and Parameter Recommendation

Chinmay Gupta · 6 November 2026 · One-page summary of the full risk memo · Fund modelled at $50,000,000 design scale on KPK's published carry parameters

An agent-based Monte Carlo engine built to stress-test KPK's BTC carry design as a risk desk would before real capital moves. The engine sweeps **1,080 parameter combinations** across three price generators (block bootstrap, jump diffusion, regime switching), drives the fund against **12 named event scenarios**, with liquidators, depositors and arbitrageurs modelled as agents. It is validated against KPK's own record: the engine replays the fund's 29 October 2026 venue-freeze unwind and reproduces it exactly (277,337 USDC repaid, 4.925 WBTC returned, zero liquidation loss). Parameters below were picked by a disclosed mechanical selection rule, not by hand.

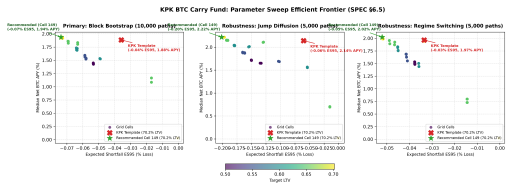
THE RECOMMENDATION

Parameter	KPK published template	Recommended (Cell 149)
Target loan-to-value	70.2%	**70.2%** (retained)
Health factor unwind floor	1.13	**1.05**
Collateral mix	100% WBTC	**70:30 WBTC to cbBTC**
Borrowed USDC deployment	KPK USDC yield vaults	**Equal thirds across three vaults**

1.94%
Median carry APY, recommended vs 1.88% template

21.6%
Unwind probability on standard paths vs 28.3% template

3.42%
Portfolio APY at 45% carry vs 3.39% template



Efficient frontier, 1,080 cells, three generator panels.

STRESS RESULTS (ES95 TAIL LOSS / LIQUIDATION PROBABILITY)

Scenario	KPK template	Recommended
S1 Flash crash (Oct 2025 cascade)	+3.39% / 4.0%	+3.39% / 4.0%
S4 Wrapper depeg (cbBTC -5%)	+4.99% / 0.0%	**+1.49% / 0.0%**
S5 Issuer exploit (Resolv pattern)	+20.59% / 0.0%	**+13.86% / 0.0%**
S6 Counterparty default (Stream pattern)	+29.72% / 0.0%	**+20.11% / 0.0%**
S12 Composite: crash + freeze + depeg	+9.00% / 65.5%	**+5.65% / 10.8%**

LIMITS, STATED PLAINLY

- **No crash protection at equal leverage.** In a pure flash crash the recommended setting loses exactly what the template loses. What it buys: fewer false-alarm unwinds and composite-stress resilience.
- **Calibration is partial.** The bootstrap generator passes every calibration check; the jump diffusion and regime switching generators miss the strict volatility-clustering tolerance.
- **Zero liquidations on standard paths is structural, and assumptions are labelled.** The unwind floor fires first on standard paths; liquidation risk shows up in the scenarios above. The AMM sleeve uses an assumed 6% yield input and the funding sleeve a labelled proxy; neither is a simulated result.

Traceability: every figure traces to a logged run in report/runs.jsonl (1,126 runs; grid stamped 7b809d8b, downstream 520262df), reproducible from the codebase. An earlier draft was withdrawn after adversarial review, disclosed in the memo. Full memo (15 pp.), factsheet, calibration note, code and run log: **github.com/Chin-mae/kpk-btc-carry-sim**